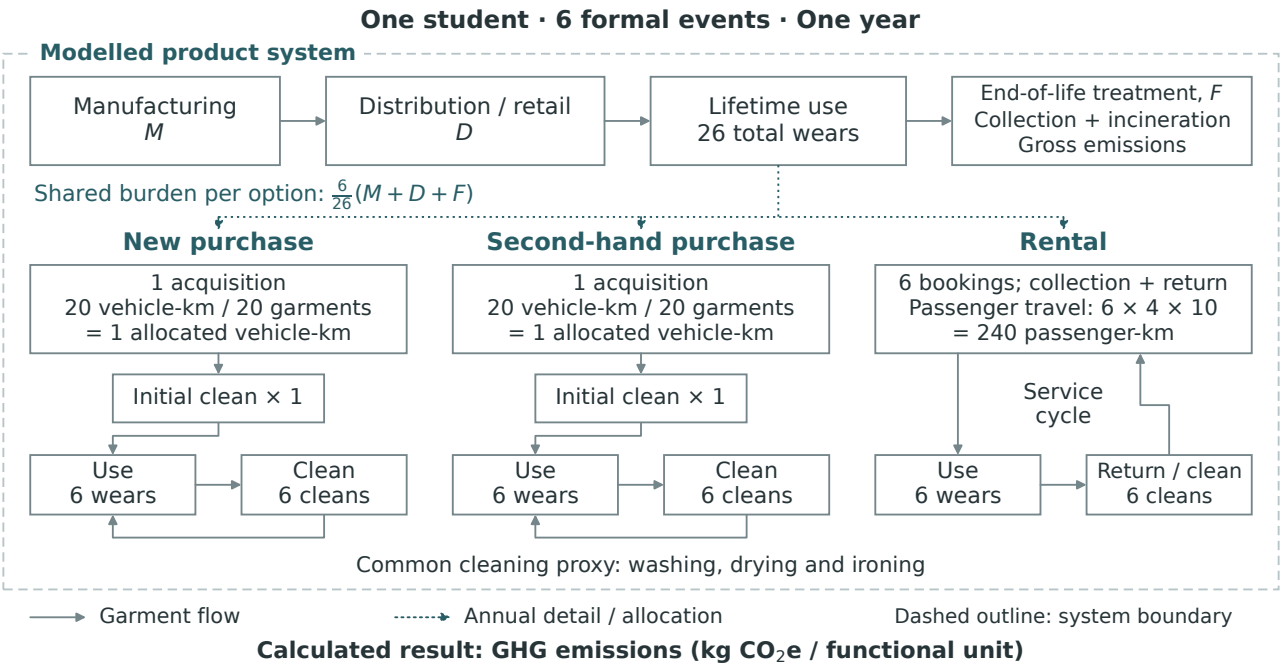
One student · 6 formal events · One year
Modelled product system
Manufacturing
M
Distribution / retail
D
Lifetime use
26 total wears
End-of-life treatment, F
Collection + incineration
Gross emissions
Shared burden per option: 6/26 (M + D + F)
New purchase
1 acquisition
20 vehicle-km / 20 garments
= 1 allocated vehicle-km
Initial clean × 1
Use
6 wears
Clean
6 cleans
Second-hand purchase
1 acquisition
20 vehicle-km / 20 garments
= 1 allocated vehicle-km
Initial clean × 1
Use
6 wears
Clean
6 cleans
Rental
6 bookings; collection + return
Passenger travel: 6 × 4 × 10
= 240 passenger-km
Service
cycle
Use
6 wears
Return / clean
6 cleans
Common cleaning proxy: washing, drying and ironing
Garment flow
Annual detail / allocation
Dashed outline: system boundary
Calculated result: GHG emissions (kg CO2e / functional unit)